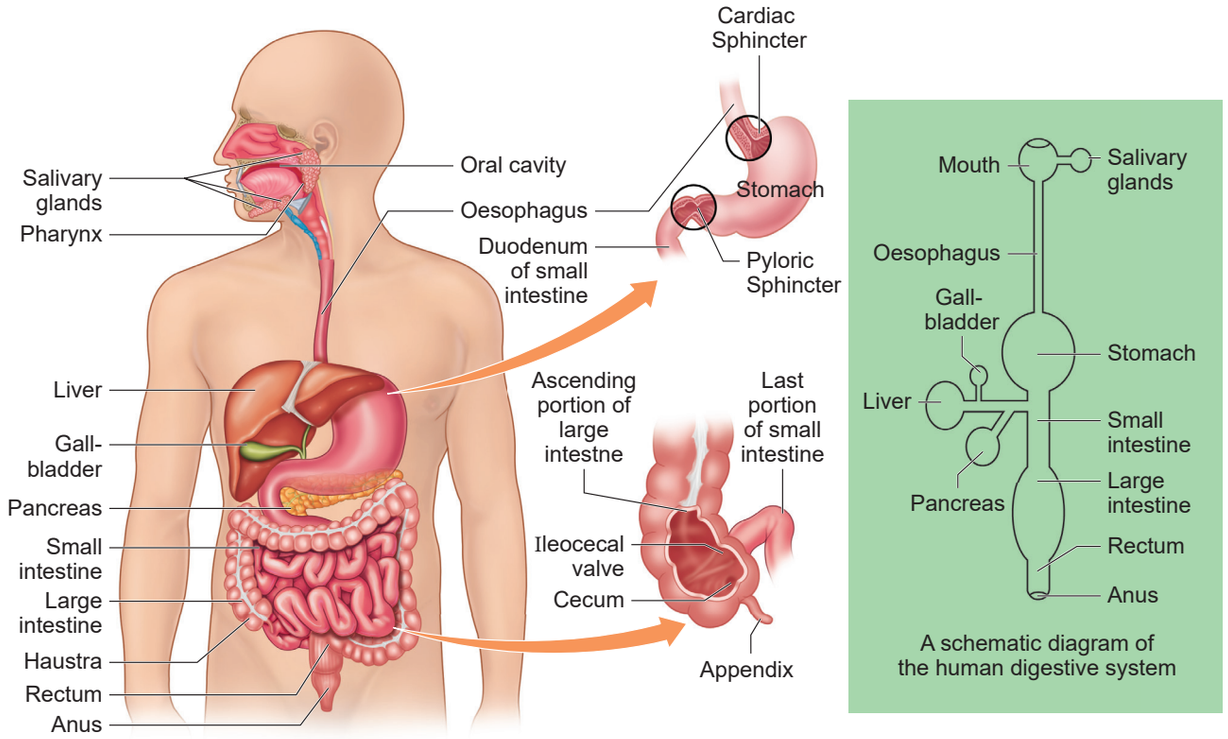


Figure 5.1 The Human Digestive system

swallowing. Two masses of lymphoid tissue called tonsils are also located at the sides of the pharynx.

Oesophagus is a thin long muscular tube concerned with conduction of the food to a 'J' shaped stomach passing through the neck, thorax and diaphragm. A cardiac sphincter (gastro oesphageal sphincter) regulates the opening of oesophagus into the stomach (Figure. 5.1). If the cardiac sphincter does not contract properly during the churning action of the stomach the gastric juice with acid may flow back into the oesophagus and cause heart burn, resulting in **GERD** (Gastero Oesophagus Reflex Disorder).

The stomach functions as the temporary storage organ for food and is located in the upper left portion of the abdominal cavity. It consists of three parts – a cardiac portion into which the oesophagus opens a fundic portion and a pyloric portion that opens into the duodenum. The opening of the stomach into the duodenum is guarded by the pyloric sphincter. It periodically allows partially digested food to enter the duodenum and also prevents regurgitation of food. The inner wall of stomach has many folds called **gastric rugae** which unfolds to accommodate a large meal.

The small intestine assists in the final digestion and absorption of food. It is the longest part of the alimentary canal and has three regions, a 'U' shaped duodenum (25cm long), a long coiled middle portion jejunum (2.4m long) and a highly coiled ileum (3.5m long). The wall of the duodenum has **Brunner's glands** which secrete mucus and enzymes. Ileum is the longest part of the small intestine and opens into the caecum of the large intestine. The ileal mucosa has numerous vascular projections called **villi** which are

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